[INCH-POUND]
GGG-C-313D
September 27, 1995
SUPERSEDING
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August 16, 1976

FEDERAL SPECIFICATION

CHISELS, HAND: CAPE, DIAMOND POINT, COLD, ROUND NOSE, AND SIDE CUTTING

The General Services Administration has authorized the use of this Federal Specification by all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers hand chisels used for cutting or shaping metals.

1.1.1 Federal specification coverage. Federal specifications do not cover all varieties of the commodity indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Types, classes, and styles. Hand chisels (metal working) shall be of the following types, classes, and styles, as specified (see 6.2):

Type I - Cape
  Style A - Normal duty
  Style B - Heavy duty

Type II - Diamond point
  Style A - Normal duty
  Style B - Heavy duty

Type IV - Cold
  Class 1 - Regular length
  Class 2 - Long

Type VI - Round nose
  Class 1 - Half
  Class 2 - Full

Type VII - Side-cutting

1.2.2 Sizes. Chisels covered by this specification shall be of the sizes shown in the respective tables, as specified (see 6.2).

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any other data which may improve this document should be sent to the General Services Administration, Tools and Appliances Commodity Center, Washington, DC 20406.

FSC 5110

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2. APPLICABLE DOCUMENTS

2.1 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American National Standard Institute (ANSI):

ANSI B46.1	-	Surface Texture; Surface Roughness; Waviness and Lay; and Surface Finish Roughness Height (in Microinches)
ANSI/HTI B209.1		Metal Chisels, Safety Requirements

(Private sector and civil agencies may purchase copies from the American National Standards Institute, 11 West 42nd Street, New York, NY 10036.)

American National Standards Institute (ANSI)/American Society of Quality Control (ASQC):

ANSI/ASQC Z1.4	-	Sampling Procedures and Tables for Inspection by Attributes

(Private sector and civil agencies may purchase copies from the American Society for Quality Control, P.O. Box 3005, 611 E. Wisconsin Avenue, Milwaukee, WI 53201-4606.)

American Society for Testing and Materials (ASTM):

ASTM A 29/A29M		Specification for Steel Bars, Carbon and Alloy, Hot Wrought and Cold-Finished, General Requirements for
ASTM E 18	-	Methods of Test for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials
ASTM E 165	-	Liquid Penetrant Inspection Method
ASTM E 709	-	Magnetic Particle Examination

(Application for copies should be addressed to the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

(DoD activities may obtain copies of those adopted voluntary standards listed in the DoD Index of Specifications and Standards free of charge from the Standardization Document Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

3. REQUIREMENTS

3.1 Illustrations. Except for the technical requirements (e.g., design, dimensions), the illustrations herein are descriptive and are not intended to preclude the purchase of chisels otherwise conforming to the requirements of this specification.

3.2 Material. Chisels shall be hot-forged of any alloy steel having good cutting, wear, and shock-resisting properties in order to withstand the tests specified in 4.5. Special quality, fine grain, hot rolled or cold finished, carbon steel may also be used. Where tables list body size cross section as a minimum, stock sizes that are undersize within commercial tolerances for bar stock as specified in ASTM A 29/A 29M are acceptable.

3.3 Hardness.

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3.3.1 Cutting end. The cutting end shall have a Rockwell C hardness of 53 to 59 when tested as specified in 4.5.

3.3.2 Struck end. The struck end shall have a Rockwell C hardness of 35 to 45 when tested as specified in 4.5.

3.4 Finish. The chisels shall be free of nonfunctional sharp edges, points, and surface roughness. The surfaces forming the cutting edge and the length (dimension E) shall have a finish with a maximum of 125 (Arithmetical Average) roughness height rating. The struck face of all chisels shall either be chamfered or provided with an upset head to minimize mushrooming (see Figure 1) and shall conform to ANSI B 209.1.

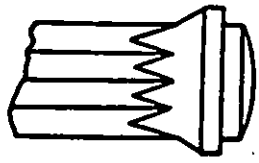


Figure 1. Optional head design for all chisels struck ends.

3.5 Body shape. The bodies shall be square, hexagonal, or octagonal in cross section. Chisels with a square body shall be either quarter octagon or have the corners rounded.

3.6 Identification marking. Unless otherwise specified, chisels shall be marked with the manufacturer's name or identifying symbol, and the state or country of manufacturer. All markings shall be engraved, etched, molded, or stamped directly on the item's surface in such a manner that it remains clearly legible throughout the life of the item. In addition, the chisels shall be marked with the safety requirements specified in ANSI/HTI B209.1.

3.7 Type I, cape. Type I chisels shall have a double-beveled cutting edge ground on both sides to an included angle (see Figure 2).

3.7.1 Style A, normal duty. Style A chisels shall conform to Table I and be similar to Figure 2.

3.7.2 Style B, heavy duty. Style B chisels shall conform to Table II and be similar to Figure 2.

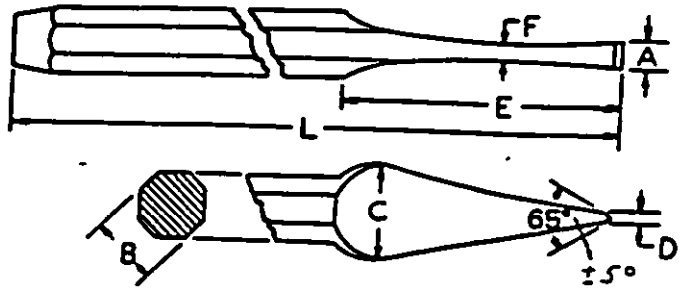


Figure 2.. Type I, styles A and B, cape, normal duty and heavy duty

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Table I. Type I, style A, cape, normal duty

Size (width of cutting edge) A ± 1/32 (inch)	Distance across flats stock size B minimum (inch)	Widest section of blade C minimum (inches)	Thickness at work end D ± 1/64 (inch)	Length E minimum (inches)	Thinnest section of blade F + 1/32, - 1/64 (inch)	Overall length L minimum (inches)
1/8	5/16	1/2	3/32	1-1/2	3/32	4-1/2
3/16	5/16	1/2	3/32	1-5/8	1/8	5
1/4	3/8	9/16	3/32	1-5/8	5/32	5
5/16	1/2	5/8	7/64	1-3/4	3/16	6
3/8	1/2	3/4	7/64	2-1/8	1/4	6-1/2
1/2	5/8	7/8	7/64	2-3/4	11/32	7
5/8	3/4	15/16	1/8	3-1/2	3/8	7
3/4	7/8	1-1/6	1/8	4	1/2	8

Table II. Type I, style B, cape, heavy duty

Size (width of cutting edge) A ± 1/32 (inch)	Distance across flats stock size B minimum (inch)	Widest section of blade C minimum (inches)	Thickness at work end D ± 1/64 (inch)	Length E minimum (inches)	Thinnest section of blade F ± 1/64 (inch)	Overall length L minimum (inches)
3/16	19/32	23/32	3/32	2	1/8	6-3/4
1/4	1/2	11/16	3/32	2-5/8	9/64	6-1/2
5/16	23/32	1-3/32	1/8	2-15/16	7/32	7-1/4

3.8 Type II, diamond point. Type II chisels shall have the cutting end tapered and square in cross section. The cutting end shall be provided with either a single or double bevel having a diamond-shaped surface.

3.8.1 Style A, normal duty. Style A chisels shall conform to Table III and be similar to Figure 3.

3.8.2 Style B, heavy duty. Style B chisels shall conform to Table IV and be similar to Figure 3.

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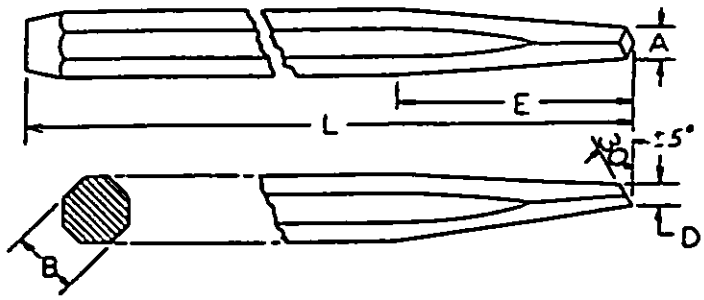


Figure 3. Type II, styles A and B, diamond point, normal and heavy duty.

Table III. Type II, style A, diamond point, normal duty

Size (width of cutting edge) A ± 1/32 (inch)	Distance across flats stock size B minimum (inch)	Thickness at work end D ± 1/32 (inches)	Length of taper E minimum (inches)	Overall length L minimum (inches)
1/8	5/16	1-1/2	1-1/2	4-1/2
1/4	3/8	1-3/4	1-3/4	5-1/2
5/16	1/2	2	2	5-3/4
3/8	1/2	2-1/8	2-1/8	6-1/4
1/2	5/8	2-3/4	2-3/4	7
5/8	3/4	3	3	7-1/2
3/4	7/8	3	3	8

Table IV. Type II, style B, diamond point, heavy duty

Size (width of cutting edge) A ± 1/32 (inch)	Distance across flats stock size B minimum (inch)	Thickness at work end D ± 1/32 (inch)	Length of taper E minimum (inches)	Overall length L minimum (inches)
11/32	3/4	11/32	2-1/4	7

3.9 Type IV, cold chisel. Type IV chisels shall have a double-beveled cutting edge ground on both sides to an included angle (see Figure 4). The cutting edge shall be straight and shall be 90° to the longitudinal axis of the chisel ± 2°.

3.9.1 Class 1, regular length. Class 1 chisels shall conform to Table V and be similar to Figure 4.

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3.9.2 Class 2, long. Class 2 chisels shall conform to Table VI and be similar to Figure 4.

3.9.3 Type IV, sets. When specified, the type IV chisels shall be furnished in a fabric or plastic wrap or roll, with ties or snap fasteners.

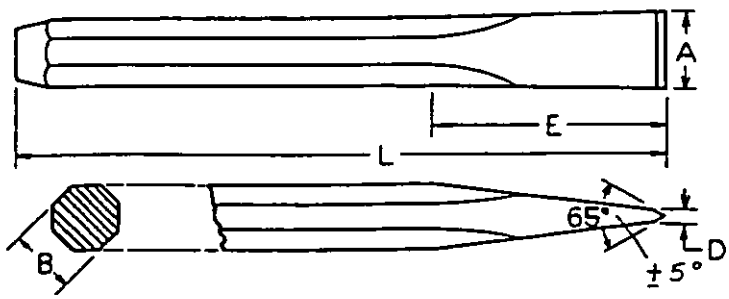


Figure 4. Type IV, cold chisel.

Table V. Type IV, class 1, regular length cold chisels

Size (width of cutting edge) A + 1/16, - 1/32 (inch)	Distance across flats stock size B minimum (inch)	Thickness at work end D + 1/32, - 1/64 (inch)	Length of taper E minimum (inches)	Overall length L minimum (inches)
1/4	3/16	1/16	1-13/32	4
5/16	1/4	1/16	1-1/2	4
3/8	5/16	1/16	1-3/4	5
7/16	3/8	1/16	2	5-1/2
1/2	7/16	3/32	2	5-3/4
5/8	1/2	3/32	2	6
3/4	5/8*	1/8	2-1/4	6-1/2
7/8	3/4	9/64	2-1/2	7-1/2
1	7/8	5/32	2-3/4	8
1-1/8	1	5/32	3-1/4	8-1/2
1-3/16	1	5/32	3-1/4	9

* Tolerance of ± 0.007

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Table VI. Type IV, class 2, long cold chisels

Size (width of cutting edge) A + 1/16, - 1/32 (inches)	Distance across flats stock size B minimum (inches)	Thickness at work end D + 1/32, - 1/64 (inch)	Length of taper E minimum (inches)	Overall length L ± 1/4 (inches)
1/2	7/16	3/32	2	10
5/8	1/2	3/32	2	11, 12
3/4	5/8	1/8	2-1/4	8, 12, 16, 18
7/8	3/4	9/64	2-1/2	12, 14, 18
1	7/8	5/32	2-3/4	12, 16, 18
1-1/4	1-1/8	5/32	3-1/4	24, 36
1-1/2	1-3/8	3/16	3-1/2	24

3.10 Type VI, round nose.

3.10.1 Class 1, half. Class 1 chisels shall have a semicircular cutting edge with a single bevel and shall conform to Table VII and be similar to Figure 5.

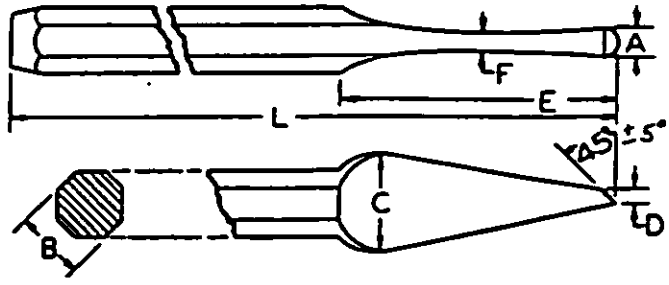


Figure 5. Type VI, class 1, half, round nose chisel, normal duty.

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Table VII. Type VI, class 1, normal duty, half round nose chisel

Size (width of cutting edge) A ± 1/32 (inch)	Distance across flats stock size B minimum (inch)	Widest section of blade C minimum (inches)	Thickness at work end D ± 1/16 (inch)	Length E minimum (inches)	Thinnest section of blade F + 1/32, - 1/64 (inch)	Overall length L minimum (inches)
1/8	5/16	7/16	.062	1-1/2	3/32	4-3/4
3/16	5/16	7/16	1/8	1-3/4	1/8	5
1/4	3/8	9/16	1/8	1-3/4	5/32	5-1/2
5/16	1/2	11/16	5/32	2-1/4	3/16	6
3/8	1/2	11/16	3/16	2-1/4	1/4	6
1/2	5/8	15/16	1/4	2-3/4	11/32	7
5/8	3/4	1	5/16	3-1/4	3/8	7-1/2
3/4	7/8	1-1/6	3/8	3-1/2	15/32	8

3.10.2 Class 2, full. Class 2 chisels shall have the work end eccentrically tapered with respect to the longitudinal axis of the chisel body. The cutting edge shall be single-bevel ground so that the cutting edge has the shortest radius of the resulting flat, measured from the longitudinal axis of the chisel body. Class 2 chisels shall conform to Table VIII and shall be similar to Figure 6.

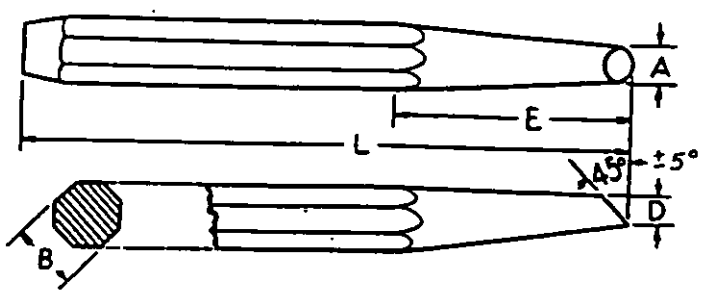


Figure 6. Type VI, class 2, full, round nose chisel.

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Table VIII. Type VI, class 2, full-round nose chisel

Size (width of cutting edge) A ± 1/32 (inch)	Distance across flats stock size B minimum (inch)	Thickness at work end D ± 1/32 (inch)	Length of taper E minimum (inches)	Overall length L minimum (inches)
1/4	3/8	9/32	2-1/4	5-1/2
5/16	1/2	11/32	2-1/2	6
1/2	3/4	17/32	3	7

3.11 Type VII, side cutting. Type VII chisels shall have a single beveled cutting edge. The cutting end shall be tapered so that the cutting edge is directly in line with the longitudinal axis of the body. Type VII chisels shall conform to Table IX and be similar to Figure 7.

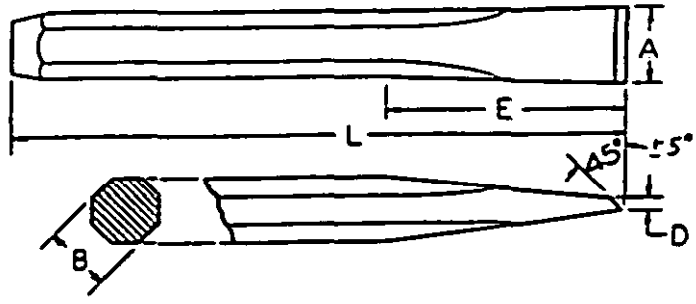


Figure 7. Type VII, side cutting chisel.

Table IX. Type VII, side cutting chisel

Size (width of cutting edge) A ± 1/32 (inch)	Distance across flats stock size B minimum (inch)	Thickness at work end D ± 1/64 (inch)	Length of taper E minimum (inches)	Overall length L minimum (inches)
7/8	23/32	9/64	2-3/4	7

3.12 Workmanship. Details of workmanship shall be in accordance with the best commercial standards and practices. The item shall be clean and free of corrosion and debris (e.g., chips, shavings, slivers) or other foreign material. The items shall be free from manufacturing workmanship defects, excepting cutting edge (e.g., sharp or rough external edges, corners, or surfaces) and material workmanship defects (e.g., pits rips, fins, burs, tears, nodules, cracks, blisters) which may adversely impact the item's serviceability, durability, safety, or appearance.

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3.13 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the tolerances specified and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the specification preparing activity for change to this document.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. In accordance with 4.1 above, the supplier is responsible for insuring that components and materials used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specification and standards to the extent specified, or, if none, in accordance with this specification.

4.2 Lot sampling procedures. Unless otherwise specified,lot sampling for inspection shall be performed in accordance with ANSI/ASQC Z1.4. The lot size shall be expressed in terms of chisels. The sample unit shall be one chisel.

4.3 Examinations.

4.3.1 Visual examinations. Each sample chisel shall be examined for design, construction, workmanship and marking. The Inspection Level shall be Level II with an Acceptable Quality Level (AQL) of 1.0 expressed in terms of defects per hundred units.

4.3.2 Dimensional examinations. Each sample chisel shall be examined for the dimensional requirements specified in the applicable paragraphs and tables. The Inspection Level shall be S-4, with an AQL of 1.0 expressed in terms of defects per hundred units.

4.4 Lot testing. Each sample chisel shall be tested in accordance with 4.5 through 4.5.2.2. The Inspection Level shall be S-4, with an AQL of 1.0 expressed in terms of percent defective.

4.5 Tests.

4.5.1 Hardness test. Chisels shall be subject to a hardness test in accordance with ASTM E 18 to determine compliance with 3.3.1 and 3.3.2. When grinding is necessary to prepare the test surface, the amount of material removed must not exceed 0.007 inches on the surface contacted by the indentor.

4.5.1.1 Cutting end. Hardness impressions on the work end of the chisels shall be taken within 1/2 inch from the cutting edge and in accordance with ASTM E 18.

4.5.1.2 Surface roughness. Surface roughness specified in 3.4 shall be tested in accordance with ANSI B 46.1.

4.5.2 Performance Test.

4.5.2.1 Impact Test. Impact testing shall be performed in accordance with ANSI B209.1.

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4.5.2.2 Magnetic particle inspection. Chisels shall be subjected to a bidirectional wet magnetic particle test in accordance with ASTM E 709 or a liquid penetrant inspection in accordance with ASTM E 165, Method A. If a chisel has one or more surface cracks, the chisel may be reworked by grounding or buffing to eliminate the surface crack(s); but must maintain the shape and dimensional requirements. All reworked chisels shall be retested. All chisels presented for acceptance shall have no surface cracks.

4.6 Examination of preparation for delivery requirements. An examination shall be made to determine that the preservation, packaging, packing, and marking comply with the requirements specified in the contract. The lot size shall be the number of shipping containers in the end item inspection lot. The sample unit shall be one container. The Inspection Level shall be S-4, with an AQL of 4.0 expressed in terms of percent defective.

Table X. Classification of preparation for delivery defects

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling Distortion of container
Contents	Number per container is more or less than required Net weight exceeds requirements

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking shall be as specified in the contract or purchase order.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. Types I, II, IV, VI, and VII chisels are intended for cutting or chipping metals. Caution: Safety goggles should be worn when using these chisels.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Type, class, and style, if applicable, of item required (see 1.2).
c. Size of chisel (width cutting end) required (see 1.2.2 and applicable tables).
d. Specify length required for type IV, class 2 chisels (see 3.9.2).

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MILITARY INTERESTS:

DoD has waived coordination on revisions and amendments to this Federal Specification until further notice.

User Activities:

Army - ME, EL, WC
Navy - SH, YD, MC
Air Force - 99

Peparing Activity:

GSA - FSS

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